

A flexible hardware accelerator for morphological filters on FPGA

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Abstract—Nowadays, the demand for embedded image processing, especially for computer vision applications is growing. Digital image processing is seeing intensive use in real time applications for several areas, such as the industrial environment, military equipment, automobile, entertainment, and medical instruments. Mathematical Morphological theory, offers a wide variety of tools, applied efficiently in most image processing steps at different scales, like image enhancement, segmentation, and classification. Therefore, such features yield a high potential of reconfiguration in image processing blocks. Morphological opening and closing operators are used as primary blocks in diverse image processing applications such as image cleaning and contrast enhancement. Moreover, using appropriate shapes and scales of structuring elements(SE), provides more flexibility and can significantly improve the enhancement performance. Nevertheless, such operators can be computationally intensive due to the increased complexity and the significant memory requirement. Also, the use of high image resolution and re-configurable size and shape of structuring elements may deepen this limitation significantly. In this paper, we consider hardware acceleration on FPGA as a promising solution to raise the above mentioned-challenges.

Both intra-level and inter-level parallelism techniques based on the reuse of optimized dilation and erosion blocs are exploited to accelerate computation and save memory consumption. Unlike most existing works, this paper proposes a highly flexible architecture for opening and closing-based morphological filters. The architecture supports re-configurable shapes and sizes of structuring elements while keeping a high ability to meet real-time processing requirements and low power consumption. An improvement in terms of frame rates up to 8% with more than 2000 Fps for moderate size of structuring elements and high image resolution is achieved.

I. INTRODUCTION

Mathematical morphology is a very rich theory that provides numerous operators used in diverse image processing applications and domains [1]. Although numerous image processing tools for image smoothing and contrast enhancement are proposed in the literature, based on morphological operations and median filters, [2], morphological based operations remain the most efficient in noise removal. Among the several mathematical morphology operations available, opening and closing operators offer challenging performances in enhancement steps. Besides, opening and closing blocks can build high efficient texture analysis.

Moreover, morphological tools can provide a high degree of flexibility, by carrying out different structuring element sizes and shapes. Such flexibility allow either multi-scale or pyramidal analysis. Commonly, the existing objects

in an input image should be treated as per the most appropriate SE to identify the desired features precisely. Hence, processing the image based on different structuring elements is very important and may increase the efficiency of the morphological algorithms in terms of flexibility and accuracy. The extracted information gives so more precise quantitative measurements of the object properties to accomplish higher-level image processing tasks like object recognition.

Nevertheless, a multi-resolution operation using a high variability of SE shapes and sizes, can require intensive computation and be time-costly. A realistic hardware environment allows using only a few specific and limited set of structuring elements. Accordingly, a reasonable trade-off between the quality of processing and computation performances should be established. In this paper, we propose a hardware accelerator for morphological enhancement and texture analysis operator on the FPGA platform. The main goal is to reduce the execution time of such blocks by reusing optimized common processing units for dilation and erosion and exploiting the inherent parallelism of morphological operators.

The remainder of this paper is organized as follows: In section 2, we detail some related works and situate our contribution. In section 3, we give preliminaries on morphological operators and detail the proposed architecture. In section 4, we present design performances implemented in Virtex-6 FPGA. Finally, we conclude and draw our perspectives in section 5.

II. RELATED WORK

A literature review on hardware acceleration systems applied to image processing, puts in evidence different aspects, while considering morphological based solutions. The first aspect consists of comparing and evaluating the efficiency and performance of different solutions and algorithms devoted to specific applications. For example, authors in [3] evaluate the suitability of embedding different stereo vision algorithms on different resource-limited systems.

A second research aspect in the literature is dealing with the adequation between application requirements and different target platforms, such as medical image processing implementation on GPUs and FPGA [4]. However, a specific platform can be efficient for some algorithms but not for others.

Another aspect of related work, consisting of hardware-oriented reviews, is focusing on the design of optimized blocks, for building particular operators, within some target

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platforms. In such issue, researchers can take advantage from the rapid growth of technology, by using new versions of platforms with better characteristics. In this work, we consider this aspect while we focus mainly on the acceleration of opening and closing based morphological filters on FPGA platform.

In this issue, two main strategies are adopted in the literature to speed up morphological computation and optimize their performances (i) software approach based on the design of efficient algorithms and (ii) hardware approach based on the design of optimized architectures.

For the software approach, the design of algorithms takes into consideration different issues:

- (a) The first category of algorithms gathers 1D and 2D algorithms. They optimize morphological computation taking into consideration the dimension of the supported structuring element. On the one hand, 1D algorithms, such as those proposed in [5], [6], are optimized only for linear structuring elements. On the other hand, 2D algorithms, such as the proposed in [6], is optimized for 2D SE.
- (b) The second category is optimized according to the level of complexity of the morphological operation. For example, some algorithms are optimized only for basic dilation and erosion, while other ones are optimized for the compound of erosion and dilation, building new complex operators namely the opening and closing morphological operators. Some proposed algorithms for opening and closing can be found in [7], [8].
- (c) The third category relies on the optimization technique at a block level: For example, the software approach relies on optimization via the decomposition of the structuring element, or via performing parallel processing on multi-core, and exploiting data parallelism. Generally, most of the existing software solutions are based on the decomposition method.

Unlike the software approach, which focuses mainly on reducing the number of comparisons per pixel, the second strategy gives more importance to hardware considerations such as parallel processing, and the regular or sequential access of data.

However, most of the existing works in the state-of-the-art which are dedicated to optimizing morphological operators are either:

- 1) Dedicated only for basic dilation and erosion.
- 2) Optimized for only limited shapes of SEs (generally for rectangular SE).
- 3) Only efficient for low image resolution processing.
- 4) Limited to process either binary or grey-level image.
- 5) Have overuse of internal memory due to the multiple-image scan or high complexity of the implemented algorithm.

Most of the proposed architectures in the literature focus only on the optimization of basic operators namely dilation, and erosion, independently of how they are used in image

processing applications. Among these architectures, delay-line architectures, data reuse architectures, and parallel and pipelined architectures can be mentioned (more detail about those architectures can be found in [9]).

Other works focus on the optimization of more complex morphological operators, which are compound operators. The latter are used in a higher level of application such as granulometry, edge detection, image smoothing. In fact, such tools require a more appropriate architecture to meet real-time requirements, low energy consumption, resources efficiency, and flexibility. In [10], Torres-Huitzil proposes a pipelined architecture for morphological granulometry computation. The architecture exploits both fine-grain parallelism and coarse-grain pipeline resulting in high-performance achievement and support of reconfigurable shapes and sizes of SEs. In [11], Kraft et al. propose a delay line-based architecture for morphological gradient calculations. The architecture achieves a high frame rate and preserves low power consumption. However, it cannot implement a fully customized size and shape. Besides, it detects edges in only four directions. In [12], Gibson et al. propose a morphological edge detection accelerator used in the augmented vision at real-time application. The architecture is characterized by its reconfigurability of the shape and sizes of SEs. However, it suffers from the same limitation of delay-line-based architecture such as the high memory requirement and latency when processing images with high resolutions or increased sizes of SEs. Recently, in [13], authors proposed a highly flexible architecture that allows extracting the standard edge gradient as well as the internal and edge gradients based on basic dilation and erosion blocs. The proposed architecture supports reconfigurable shapes and sizes of structuring elements. In [10], Bartovsky et al. propose an efficient architecture for open and close-based filters. The architecture reuses optimized modules for dilation and erosion and performs open-close calculations in four pipelined stages. The architecture achieves low latency and memory use but supports only rectangular shapes of SEs. In [14], authors propose a pipelined architecture for morphological opening closing based filters limited for rectangular structuring elements. The architecture achieves low memory utilization as it allows processing data on-stream. Also, it processes pixels belonging to adjacent windows in a parallel way. The execution time is thus reduced significantly.

Although the existing implementations of morphological opening and closing offers correct operations. They still suffer from lack of flexibility with respect to the structuring element shape and can be further improved in both computation time and architecture reconfiguration.

This paper aims to accelerate compound morphological opening and closing-based filters, while keeping a high degree of flexibility in treatment level and low timing and resource utilization.

III. THE PROPOSED DESIGN OF MORPHOLOGICAL OPENING-CLOSING BASED FILTERS

A. Preliminaries on morphological opening and closing based filters

Mathematical morphology is defined as the science of transforming images. It is applied mainly to transform the original image into another image using a window with an appropriate shape and size. All morphological applications are based on the two main basic operators dilation and erosion.

The dilation and erosion of an image I by a structuring element SE at position (h, n) are denoted by $(I \oplus SE)$ and $(I \ominus SE)$ respectively, and defined by:

$$(I \oplus SE)(h, n) = \max\{I(h - a, n - b) | (a, b) \in SE\} \quad (1)$$

$$(I \ominus SE)(h, n) = \min\{I(h + a, n + b) | (a, b) \in SE\} \quad (2)$$

By concatenating the basic dilation and erosion, it is possible to obtain two important higher-order operators, which are the opening and the closing, also called morphological filters. The main difference between those operators is in the order in which the dilation and erosion are applied. They are used to remove specific details while preserving at the same time the shape and structure of existing objects in the image. They are frequently used as building blocks for other image processing tasks and more complex morphological algorithms.

The opening is performed by first eroding the input image. The eroded image is then dilated with the same structuring element. The opening operation is defined as follows :

$$\text{Opening} : O(I) = I \circ SE = (I \ominus SE) \oplus SE \quad (3)$$

In a similar way, closing is performed by first dilating the input image. The dilated image is then eroded by using the same structuring element SE .

The closing operation is defined as the following:

$$\text{Closing} : C(I) = I \bullet SE = (I \oplus SE) \ominus SE \quad (4)$$

Optimizing morphological opening and closing have great importance in the real-image processing field as they are used in diverse applications such as: image smoothing filters, granulometry, alternate sequential filter (ASF), black-hat, and white-hat operators.

B. Morphological opening and closing based Processing Units: PUOC

In this section, we present our pipelined architecture composed of opening and/or closing blocs (PUOC) designed to accelerate the basic morphological opening or

closing (Open/Close) as well as the compound filters, namely the opening of closing or the closing of opening operations (Open-Close). The index "n" in $(PUOC)_n$ defines the number of erosion and dilation stages constituting the architectures. As a reminder, a morphological open-close is defined as a concatenation of two dilations and two erosions.

The proposed architecture is illustrated in Figure.1. In this figure, we take as example $n=2$ ($(PUOC)_2$). The architecture is composed of two identical arrays of elementary processing units dedicated for basic dilation and erosion shown in Figure.1. The architecture of PUDE is more detailed in our previous work [9]. Similarly, figure.2 illustrates the architecture $(PUOC)_4$ ($n=4$) proposed for morphological sequential opening and closing. The register (Reg_inter) is used to hold the intermediate result obtained from the previous stage of dilation or erosion. (Reg_inter) is an 8-bit register (It is a 1-bit register in the case of binary image). The stored value in (Reg_inter) is used as input data to the next stage. Here, the key idea for optimization is to exploit the inter-level parallelism, which allows the simultaneous execution of atomic dilation and erosion at different neighboring data. Thus, computation is accelerated significantly.

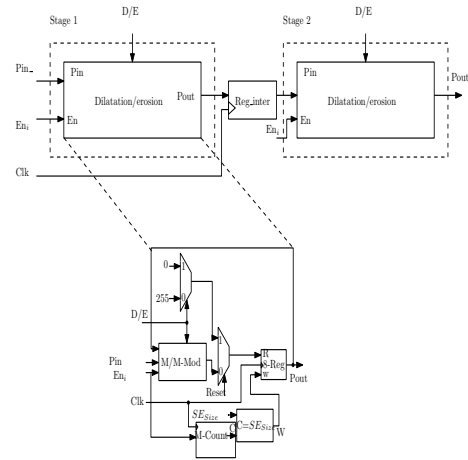


Fig. 1. Two pipelined stages based Processing Unit for basic Morphological Opening and Closing (PUOC)₂.

Figure.3 present the timing diagram of the two pipelined stages architecture proposed for morphological basic opening and closing (open/close). As shown, the execution time is reduced significantly due to the use of inter-level parallelism. Indeed, the execution of one stage begins when the first result from the previous stage is generated. For comparison purposes, the naive sequential execution takes a total time of $(2 \times T_{DE})$ (CF. figure.3).

C. Array of pipelined morphological processing units

Elementary processing units are duplicated and organized under an array for accelerating the different morphological operations. Figure.5 illustrates an example of a network of pipelined elementary processing units, that can be either opening or/and closing units (PUOC). The different cores are described and illustrated previously in Figure.1, and Figure.2.

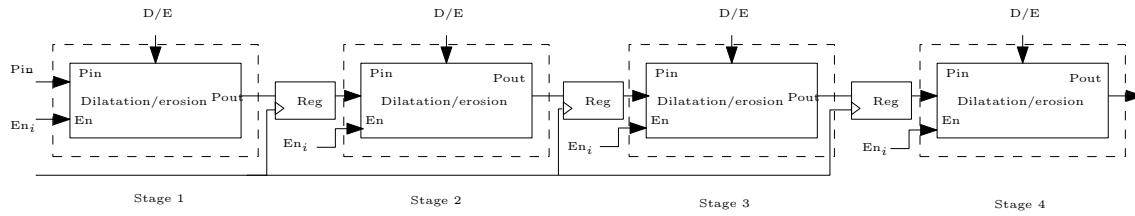


Fig. 2. Four pipelined stages based Processing Unit for morphological sequential filters, namely the Opening of Closing or Closing of Opening (PUOC)₄.

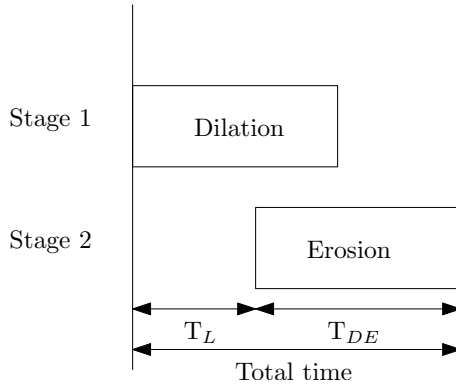


Fig. 3. The timing diagram of (PUOC)₂ architecture.

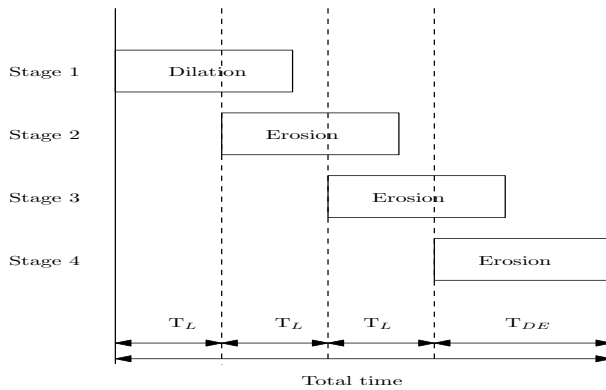


Fig. 4. The timing diagram of (PUOC)₄ architecture.

Here, the dimension and the size of the array are equal to those of the selected SE. For example, the number of processing units corresponding to 5×5 SE is 25.

The final results obtained from the set of processing units in the array are sent to an output memory using DataR. DataR is a multiplexer used to collect and generate the output which is the final results of processing. The orchestration and control of the multiplexer are ensured by the control unit. The output of an appropriate pixel is ensured using (Sel) when the result is ready. The stream of data is broadcast for all atomic cores belonging to the array, one pixel per clock cycle. Each atomic PUOC is activated and deactivated at a specific time using signals generated by the enable signal generator to select its appropriate neighboring pixels (more detail about enable signal generator in our previous work [13]).

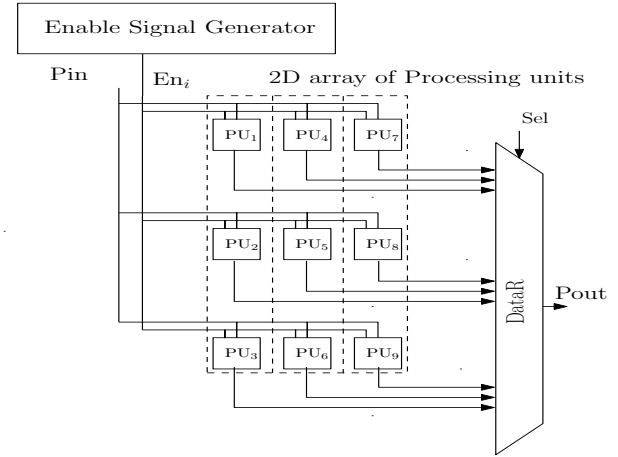


Fig. 5. Example of an Array of 3×3 of Processing Units.

IV. SYNTHESIS RESULTS AND PERFORMANCE EVALUATION OF THE PROPOSED MORPHOLOGICAL ARCHITECTURES

This section is devoted to evaluate the performance of the proposed designs. The evaluation is based on a set of measures used mainly to estimate the performances of the hardware cores in terms of execution time and FPGA resource occupation. The choice of FPGA as the target device is justified by the amenability of the morphological operators to the pipeline and parallelism. Moreover, FPGA presents one of the best solutions to create hardware accelerators with high performances [15]. The adopted framework is ISE 14.2 Design Suite tool from XILINX. The chosen language for description is the Very High-Speed Integrated Circuit Hardware Description Language (VHDL). Results consider the scale of the design with diverse parameters such as the properties of the structuring element and input image. More precisely, two-timing benchmarks are exploited to evaluate performances. The first one is against different sizes and shapes of SE, and the second one is against various image resolutions. We also illustrate the result of some processed images. Besides, we compare our results with previous works proposed for hardware acceleration of similar morphological operators in the state-of-the-art. We also compare the performances of the proposed hardware cores for cleaning filters with other hardware cores based on different operators. The proposed architectures have been simulated, synthesized, placed, and routed successfully on different FPGA XILINX platforms such as Spartan3, Virtex5, and Virtex6. For simplification, we

only illustrate results implemented on Virtex6-xc6vlx75t. In some cases, we report implementation results on other FPGA platforms to keep a fair comparison by using similar targets of prior works in the state-of-the-art.

A. Performances evaluation

Table.I illustrates the performance evaluation and resource utilization of the proposed architecture performing morphological basic opening and closing (open-close). Similarly, table.II illustrates the performances of the compound operators: opening of closing and closing of opening cleaning filters (open/close), respectively. It is clear from the two tables that the total number of slice registers and slice LUTs increases relatively with the increase of SE size and the increase of stages' number. For example, in the case of either basic opening or closing, which is based on one dilation stage and another erosion stage, the architecture occupies only 1% of slice LUTs and slice registers when processed with a small 3×3 SE, while it occupies 3% of slice LUTs and 9% of slice registers for large size SE of 11×11 . As for the compound opening of closing (or closing of opening) cleaning filters, it occupies 2% of slice registers and 7% of slice LUTs using a 3×3 SE, and 5% of slice registers and 22% of slice LUTs when using a 7×7 SE.

From Table.I and Table.II, results show also that the dynamic power consumption remains relatively low even for a moderate size of SE. The proposed architecture achieves high throughput. For example, it is about 197.18 and 196.9 MgPixel/s when using 3×3 SE in (open/close) and (open-close) based architectures, respectively. Our design can process a high frame rate. For example, the minimum frame rate is equal to 1830 and 1737 while using a large 11×11 SE in (open/close) and (open-close) based architectures, respectively.

Figure.6 depicts the minimum frame rate to perform both basic and compound morphological cleaning filters. Results are generated for different image sizes and an 11×11 moderate SE. It is clear from the image that the frame rate increases proportionally with the decrease of the input image resolution. However, the obtained results still much greater than the standard real-time frame rate (is equal to 30 frames per second) even for high image resolution. For example, the proposed architecture achieves a frame rate which is equal to 155 fps and 146 fps, and execution time of about 6.46 ms and 6.8 ms when processing one 1024×768 image based on two and four pipelined stages, respectively.

B. Comparative study with existing implementations

To proceed to a fair comparison with the related work, the proposed architecture designed for morphological filtering is compared with architectures proposed in [14], [16]. Table.III summarizes their performance evaluation for a 566×566 input image, 3×3 SE, and 7×7 SE, within the same FPGA target platform (Virtex-5). It is clear from the table that the results are very close to each other. Our architecture achieves better performances in terms of computational speed and flexibility for a 3×3 compared to the architecture in

[14] with an increase of 8% in the frame rate. However, the performance of the proposed design in [14] outperforms slightly that of our proposed design when applying a larger 7×7 SE. This degradation is due to the added time for synchronization between input data and the enable signal when processing the strip. The accumulation of synchronization times becomes more important and results in slight performance degradation, especially when applying a large size of structuring elements under multiple stages. However, our proposed architecture allows selecting any shape and size of SE, while architecture in [14] is limited to only rectangular SE, which limits its use in a large number of applications.

V. CONCLUSION

In this paper, an optimized flexible architecture is proposed for basic morphological opening and closing, their conjunction for building higher level blocks such as morphological cleaning. The architecture is intended for efficient use either in image enhancement tasks such as image smoothing and contrast enhancement applications, or in feature extraction steps. Moreover, such blocs can be exploited in more complex morphological processing, such as sequential filters, granulometry, etc. These architectures support any shape and size of structuring elements, providing a high degree of flexibility in processing level. They exploit both intra and inter-level parallelism and allow a reuse of optimized blocs for basic dilation and erosion, which accelerates efficient computations and keeps low resource utilization. The proposed designs keep good performances for high image resolution and large structuring elements. In future work, we intend for integrating such challenging blocks in high level image processing applications.

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TABLE I

RESOURCE USE AND PERFORMANCE EVALUATION FOR A 256×256 IMAGE AND DIFFERENT SIZES OF SE ON VIRTEX-6-XC6VLX75T FPGA OF MORPHOLOGICAL OPENING, CLOSING.

	Slice Reg	Slice LUTs	Dynamic power (W)	Latency (Clock cycles)	Frequency (MHz)	Throughput (Mpx/s)	Frame rate (Fps)
3×3	206(1%)	268(1%)	0.076	50	331.2	197.18	3020
7×7	1197(1%)	1505(3%)	0.123	238	266.325	140.12	2172
11×11	3057(3%)	4250(9%)	0.226	882	235.192	118.56	1830

TABLE II

RESOURCE USE AND PERFORMANCE EVALUATION FOR A 256×256 IMAGE AND DIFFERENT SIZES OF SE ON VIRTEX-6-XC6VLX75T FPGA OF COMPOUND MORPHOLOGICAL OPENING OF CLOSING FILTERS BASED ON (OPEN-CLOSE) CORES.

	Slice Reg	Slice LUTs	Dynamic power(W)	Latency (Clk cycles)	Frequency (MHz)	Throughput (Mpx/s)	Frame rate(Fps)
3×3	350(1%)	602 (1%)	0.097	100	330.64	196.9	3015
7×7	1980(2%)	3660 (7%)	0.224	676	250.125	131.59	2039
11×11	4993(5%)	10479(22%)	0.513	1764	223.229	112.533	1737

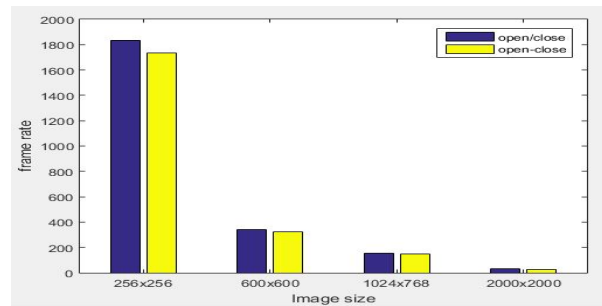


Fig. 6. Frame rate (Fps) for basic and compound opening and closing cleaning filters for different frame sizes and a 11×11 SE.

TABLE III

PERFORMANCE COMPARISON BETWEEN ARCHITECTURES DEDICATED FOR MORPHOLOGICAL FILTER FOR A 566×566 INPUT IMAGE ON VIRTEX-5 FPGA AND 3×3 AND 7×7 SEs.

	[14]	Proposed work	[14]	Proposed work
Architecture type	Parallel	Pipelined	Parallel	pipelined
supported SE	3×3	3×3	7×7	7×7
	Rectangle	Arbitrary	Rectangle	Arbitrary
Max Frequency (Mhz)	190.223	330.64	196.967	250.125
Latency (Clk)	4288	100	14564	676
Frame rate (Fps)	573	619	614	420
Throughput(Mpx/S)	190.223	196.9	196.967	131.59

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